

# test-coverage

## LLMProvider Test Coverage Ledger (round-276)

This ledger maps the surface-area symbols of LLMProvider to the specific tests + Challenges that exercise them with positive runtime evidence (CONST-035 / Article XI §11.9, CONST-050(B)).

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): "all existing tests and Challenges do work in anti-bluff manner

- they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

### Anti-bluff guarantees

Every row below identifies BOTH a symbol AND the failing-line a mutation would produce. Mutation pairing per §1.1 / CONST-050(A).

### Ledger — pkg/circuit

| Symbol                        | Test                                    | Challenge invariant               |
|-------------------------------|-----------------------------------------|-----------------------------------|
| NewCircuitBreaker()           | pkg/circuit/<br>circuit_breaker_test.go | runner construction p             |
| NewDefaultCircuitBreaker()    | pkg/circuit/<br>circuit_breaker_test.go | (covered via NewCircu             |
| DefaultCircuitBreakerConfig() | pkg/circuit/<br>circuit_breaker_test.go | runner uses defaults o<br>fixture |
| CircuitBreaker.GetState()     |                                         | circuit.initial_state.            |

| Symbol                                 | Test                                    | Challenge invariant                       |
|----------------------------------------|-----------------------------------------|-------------------------------------------|
|                                        | pkg/circuit/<br>circuit_breaker_test.go |                                           |
| CircuitBreaker.IsClosed()/<br>IsOpen() | pkg/circuit/<br>circuit_breaker_test.go | circuit.is_closed.<loc                    |
| CircuitBreaker.Complete()              | pkg/circuit/<br>circuit_breaker_test.go | circuit.opens_after_fa                    |
| CircuitBreaker.ErrCircuitOpen          | pkg/circuit/<br>circuit_breaker_test.go | runner asserts errors.<br>ErrCircuitOpen) |
| CircuitBreaker.AddListener()           | pkg/circuit/<br>circuit_breaker_test.go | (covered by unit suite                    |
| CircuitBreakerManager.Register()       | pkg/circuit/<br>circuit_breaker_test.go | (covered by unit suite                    |

## Ledger — pkg/health

| Symbol                                     | Test                                  | Challenge invariant                    |
|--------------------------------------------|---------------------------------------|----------------------------------------|
| NewHealthMonitor() / Default*              | pkg/health/<br>health_monitor_test.go | runner constructs one<br>monitor       |
| HealthMonitor.RegisterProvider()           | pkg/health/<br>health_monitor_test.go | health.initial_status.<<br>(x5)        |
| HealthMonitor.GetHealth()                  | pkg/health/<br>health_monitor_test.go | runner asserts<br>Status==Unknown init |
| HealthMonitor.RecordFailure()              | pkg/health/<br>health_monitor_test.go | health.transitions_after               |
| HealthMonitor.AddListener()                | pkg/health/<br>health_monitor_test.go | runner counts listener                 |
| HealthStatusUnknown/Unhealthy<br>constants | pkg/health/<br>health_monitor_test.go | runner asserts status s<br>als         |

## Ledger — pkg/retry

| Symbol                  | Test                        | Challenge invariant                              |
|-------------------------|-----------------------------|--------------------------------------------------|
| IsRetryableStatusCode() | pkg/retry/<br>retry_test.go | retry.is_retryable_status_code                   |
| CalculateBackoff()      | pkg/retry/<br>retry_test.go | retry.calculate_backoff_bounds                   |
| DefaultRetryConfig()    | pkg/retry/<br>retry_test.go | runner builds custom config<br>to bound MaxDelay |
| ExecuteWithRetry()      | pkg/retry/<br>retry_test.go | (covered by unit + HTTP<br>suites)               |

| Symbol                    | Test                    | Challenge invariant                            |
|---------------------------|-------------------------|------------------------------------------------|
| RetryConfig struct fields | pkg/retry/retry_test.go | runner validates Initial/Max/Multiplier bounds |

## Ledger — pkg/provider + pkg/models

| Symbol                      | Test                     | Challenge invariant                          |
|-----------------------------|--------------------------|----------------------------------------------|
| provider.LLMProvider iface  | (every provider package) | runner satisfies it via controllableProvider |
| models.LLMRequest           | pkg/models/types_test.go | runner builds real requests per-fixture      |
| models.LLMResponse          | pkg/models/types_test.go | runner threads through breaker               |
| models.ProviderCapabilities | pkg/models/types_test.go | runner asserts non-nil from controllable     |

## Challenges (challenges/ + challenges/scripts/)

| Challenge                            | Purpose                      | Mutation-paired?          |
|--------------------------------------|------------------------------|---------------------------|
| llmprovider_describe_challenge.sh    | runner invariants (23 PASS)  | <b>YES</b><br>(round-276) |
| chaos_failure_injection_challenge.sh | runtime chaos resilience     | env-gated                 |
| ddos_health_flood_challenge.sh       | DDoS health-flood resilience | env-gated                 |
| scaling_horizontal_challenge.sh      | horizontal scale check       | env-gated                 |
| stress_sustained_load_challenge.sh   | stress floor                 | env-gated                 |
| ui_terminal_interaction_challenge.sh | TUI smoke                    | env-gated                 |
| ux_end_to_end_flow_challenge.sh      | UX end-to-end                | env-gated                 |
| host_no_auto_suspend_challenge.sh    | CONST-033 host-power ban     | n                         |
| no_suspend_calls_challenge.sh        |                              | n                         |

| Challenge | Purpose                        | Mutation-paired? |
|-----------|--------------------------------|------------------|
|           | CONST-033<br>anti-suspend scan |                  |

## How to run

```
# Race-detector full unit + runner pass:
GOMAXPROCS=2 go test -count=1 -race -p 1 ./...

# Anti-bluff describe Challenge (23 invariants):
./challenges/llmprovider_describe_challenge.sh normal # exit 0
./challenges/llmprovider_describe_challenge.sh mutate # exit 99

# Standalone runner (useful for forensics):
go run ./challenges/runner/
LLMPROVIDER_MUTATE_RUNNER=1 go run ./challenges/runner/
```

## Cascade

CONST-047 + CONST-051(A): this ledger is the LLMProvider-side realisation of the cross-org "equal codebase" mandate. The same ledger pattern lives in sibling submodules (LeakHub round-266, MCP\_Module round-267, Models round-268, Ouroboros round-269, Planning round-270, conversation round-271, DebateOrchestrator round-272, HelixSpecifier round-273) and is bumped from the meta-repo on every round close-out.